

Learning System 1: Intro to Numpy and Matplotlib

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Introduction

This lab introduces fundamental concepts of Python libraries NumPy and Matplotlib, which are essential for data manipulation and visualization in machine learning. Through this lab, we learned how to create vectors and matrices and perform mathematical operations on them, followed by some linear algebra applications, such as matrix multiplication and the dot product. In the end, we learn how to visualize the given data set using Matplotlib.

Key Concepts and Methods

1. **Vector:** A vector is a one-dimensional array that represents a list of numbers.
 - **Application:** Vectors in Machine Learning usually represent data points or weights in models.
2. **Matrix:** A matrix is a two-dimensional array of numbers arranged in rows and columns.
 - **Application:** It represents an entire dataset and shows the relationships between variables.
3. **Arrays:** Arrays are data structures provided by NumPy that can store multi-dimensional data, such as vectors (1D), matrices (2D), or higher-dimensional data.
4. **Subset of Arrays:** A subset of an array refers to selecting specific rows, columns, or elements from an array based on indices or conditions
5. **Dot Product:** The dot product is a scalar value obtained by multiplying corresponding elements of two vectors and summing the results.
 - **Methodology:** In the lab task, to calculate the dot product between vector $\mathbf{u}$ and $\mathbf{v}$, we used `np.dot(u, v)`.
 - **Application:** Dot product is used to calculate the similarity between elements. For instance, dot product is used in determining if an email is "Spam" or not based on the frequency of specific keywords. Similarly, Cosine similarity using dot product is used behind Recommendation Systems like Netflix.
6. **Product of matrix and a Vector:** The product of a matrix and a vector involves taking the dot product of each row of the matrix with the vector, resulting in a new vector.
 - **Methodology:** In the lab task, to calculate the dot product between vector $\mathbf{v}$ and matrix $\mathbf{A}$, we used `A@v`.

- **Application:** This operation helps in linear transformations, and the prediction of learning models.
7. **Product of two matrices:** The product of two matrices involves computing the dot product of rows from the first matrix with columns of the second matrix, resulting in a new matrix.
- **Methodology:** In the lab task, to calculate the dot product between matrix A and matrix B, we used $A @ B$.
 - **Application:** This helps in deep learning efficiency by allowing a model to calculate the results for thousands of data samples, against all the model's weights.
8. **Prediction with a linear ML Model:**

- (a) **One output with one model:** It is used in predicting a single output for a single data point using a linear model. The prediction is computed as the dot product of the parameter vector θ and the input vector x . Mathematically,

$$h_{\theta}(x) = \theta^T x = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_d x_d$$

- **Application:** This kind of the prediction model is used in simple regression tasks where the goal is to predict single value for instance estimating the house price, temperature etc
- (b) **Multiple Outputs with one model:** It is used in predicting outputs for multiple data points using the same linear model. The predictions are computed as the matrix-vector product of the dataset matrix X and the parameter vector θ . Mathematically,

$$H_{\theta}(X) = X\theta$$

- **Application:** It is used in batch predictions where the same model is applied to multiple data points simultaneously. For instance, predicting the house price for multiple houses.
- (c) **Multiple outputs with multiple models:** It is used in predicting multiple outputs for multiple data points using multiple linear models. The predictions are computed as the matrix-matrix product of the dataset matrix X and the parameter matrix H , where each column of H is one model parameters. Mathematically,

$$Y = XH$$

- **Application:** It is used in multi-output regression tasks, where multiple outputs are predicted for each data point, for example, predicting multiple weather parameters like temperature, humidity and wind speed for a given location.
9. **Data Visualization:** We learned to construct scatter plots via matplotlib's pyplot as shown in figure 1a. These are useful for visualizing the data in the dataset when the number of features allows it, as it did for our 2D case, with the dataset containing 2 features and 300 data-points. Furthermore, we learned to apply various cosmetic enchantments to our plot, helping to highlight perceived clusters by assigning symbols, colors, and text to each slice of data (nicely aligning with the 3 clusters in our case) and is shown in Figs. 1b and 1c.

Observation

Here, we can see that the first 100 data with blue cluster have high feature 1 value and widely spread across feature 0.

For the second red cluster i.e., from 100 to 200 rows, have low values for features 1 and moderate for feature 0 resulting in most compact cluster.

Moreover, for the third green cluster i.e., from 200 to 300 rows, have moderate for feature 1 and high values for feature 0. Also, as feature 0 increases, feature 1 tends to increase slightly.

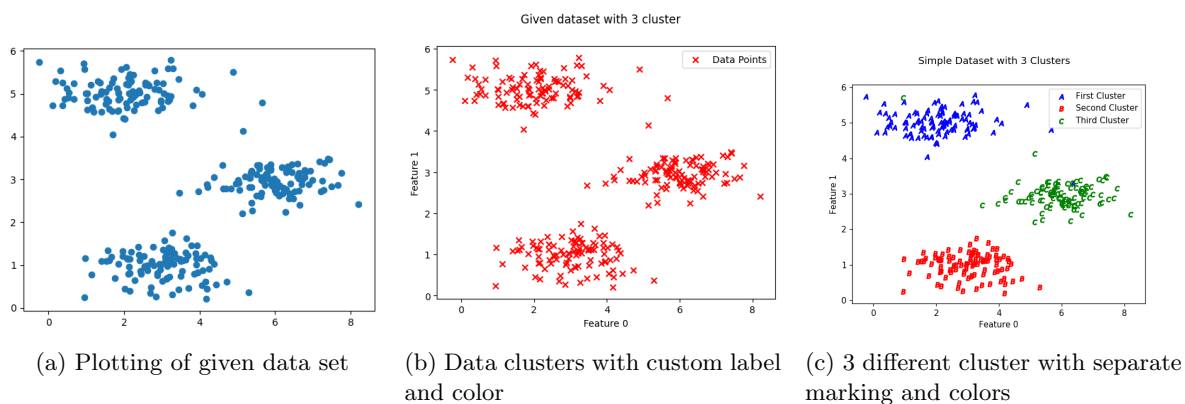


Figure 1: Three images in a single row

Conclusion

This lab served as a structured and easy to follow introduction to both Python notebook, and the concepts required for machine learning. It helped us freshen up on the mathematics needed, the syntax to perform those operations in Python code, connection between the math and machine learning, and finally, how we can grasp the data we're working with better by visualizing it.